

Mondial.eco — Buyer Journey Complete Reference

Purpose: Order থেকে payment complete পর্যন্ত পুরো system কীভাবে কাজ করে — এক জন buyer (Creator/Entrepreneur/Investor role-এ থাকা) হিসেবে step-by-step।

Traced from actual code: `WorkroomService.cs`, `LeadsService.cs`, frontend order flow + engagement components (post commit `7bd11ca`, 2026-08-02)।

পুরো journey ২টা major branch-এ যায়:

- **Auto-accept path** — instant order enabled, সব confirmation-এ pass
- **Manual approval path** — provider এর approval দরকার

দুইটাই merge হয়ে একই workroom flow-এ ঢোকে, তাই phases 3+ common।

Phase 0 — Prerequisites

Buyer-এর account থাকতে হবে (Creator/Entrepreneur/Investor role, Service Provider না)। Onboarding phase কমপক্ষে 1 হতে হবে।

Buyer login-এ থাকা অবস্থায় dashboard-এ ঢোকে।

Phase 1 — Marketplace browse

URL: `/marketplace/services`

Buyer কী দেখে:

- Left sidebar-এ filter (Category, Price range, Delivery time)
- 3-column grid-এ services
- প্রতিটা card-এ: cover image, provider name (verified chip সহ), service title, category, "From €X · N days"
- Top-এ search bar
- URL-এ deep-link support (`?category=Design&priceRange=Under $500`)

Buyer কী করে: একটা service card-এ click করে।

Screen change: listing detail page-এ redirect।

Phase 2 — Listing detail + order start

URL: `/marketplace/services/[listingId]`

Buyer কী দেখে:

- Header: category chip + service title + provider row (avatar, verified badge)
- Left column:
 - MediaCarousel (cover video/gallery)
 - Tags

- "About this service" (description)
- **Compare packages table** (Basic/Standard/Premium side-by-side, ✓/X per feature, Choose button per column)
- FAQ accordion
- "About the provider" card (bottom)
- Right sticky column:
 - Package selector (Basic/Standard/Premium tabs, Standard-এ "Recommended" pill)
 - Price + delivery + revisions
 - "What's included" list (✓ included green, X excluded muted)
 - Add-ons checkboxes (checkbox + name + (+€X))
 - Total row
 - **"Continue (€X)" button** (this is the order start)
 - "Message provider" button (stub, chat not built yet)

Buyer কী করে:

1. Tier select (Basic/Standard/Premium)
2. Optional add-ons check
3. "Continue (€X)" click

Backend action: কিছু না এখনো — frontend router push।

URL change: `/marketplace/services/[listingId]/order?step=1&packageId={pkgId}&addons=addon1,addon2`

Phase 3 — Order wizard 3-step (buyer side)

URL pattern: `/marketplace/services/[listingId]/order?step={1|2|3}&packageId=...&addons=...`

3-step wizard, sticky right-column order summary throughout।

Step 1 — Review

Buyer কী দেখে:

- Left card: "Review your order"
 - Package title, description
 - Selected add-ons list
 - Included features (green checks)
 - "🕒 Estimated delivery: N days"
- Right sticky summary: cover thumbnail, provider name, service title, tier badge, line items (base + add-ons), Total, "Provider receives €X after 12% Mondial platform fee" info box

- Footer: "Continue to requirements" button

Buyer কী করে: Continue click।

Step 2 — Requirements

Buyer কী দেখে:

- Left card: "Provider requirements — Answer the questions..."
- Fields from `(pkg.requirementsTemplate)` (text/number/date/boolean/ choice)
- Required fields-এ red asterisk
- Footer: Back + "Continue to confirmation" (disabled until required fields filled)

যদি package-এ কোনো requirement না থাকে: "No requirements needed for this package" message + Back + Continue।

Buyer কী করে: required fields fill + Continue।

Step 3 — Confirm + place order

Buyer কী দেখে:

- Left card: "Place your order"
- 3 checkboxes:
 1. "I have reviewed the final summary" (reviewedSummary)
 2. "I confirm this order" (explicitlyConfirmed)
 3. "No outstanding compliance issues" (noComplianceHold)
- Payment simulation notice: "This is a development environment. No real payment is processed; escrow authorization is simulated."
- Right summary: same as Step 1
- Footer: Back + "Place order · €X" (disabled until all 3 checked)

Buyer কী করে: ৩টা checkbox check + "Place order · €X" click।

Phase 4 — Backend PurchasePackageAsync (decision moment)

Endpoint: `(POST /api/leads/package-purchases)`

Backend কী করে (LeadsService.cs:290):

1. Package + listing fetch করে
2. Provider + client verify করে (ServiceProviderProfile থাকতে হবে provider-এর)
3. Selected add-ons snapshot নেয়
4. Requirements complete কিনা check করে
5. **10টা failure condition check করে:**
 - Listing status = Published?
 - Package status = Published?
 - InstantOrderEnabled && !ManualApprovalRequired?

- Provider verified + available?
- MaximumActiveOrders ছুঁয়েনি?
- Client onboarding phase ≥ 1 ?
- Requirements complete?
- PaymentMethodVerified?
- ExplicitlyConfirmed?
- EscrowAuthorized?
- !ComplianceHold?
- FinalSummaryShown?

Decision:

- `failures.Count == 0` → `auto = true` → Path A (Auto-accept)
- `failures.Count > 0` → `auto = false` → Path B (Manual approval)

Both paths-এ Proposal record create হয়:

- `Status = Accepted` (auto) বা `Submitted` (manual)
- `AcceptanceMode = RuleBasedInstantOrder` (auto) বা `ManualClientAcceptance` (manual)
- `EscrowStatus = Authorized` (auto) বা `AuthorizationPending` (manual)
- `PurchaseSnapshot` জমা হয় (immutable record of purchase context)

Auto path-এ: Background job enqueue —

`WorkroomConversionJob.ConvertAsync(proposalId)` | **Manual path-এ:** provider-কে notification — "Provider approval required" |

Phase 5A — Auto-accept path (buyer polling)

Buyer কী দেখে:

- Result panel: "Order accepted" (green CheckCircle icon)
- "Your payment is authorised in escrow and the provider has been notified. We're setting up your workroom..."
- Small spinner: "Setting up your workroom..."
- Polling starts (`useProposalConversionPoll`), 15 attempts, ~30 sec)

Backend background job (`WorkroomService.ConvertProposalAsync`):

1. Proposal check করে: Status=Accepted, ConversionStatus=AwaitingModule4
2. Milestone plan থেকে WorkroomMilestone list বানায় (single default milestone if plan empty)
3. Contract create করে (ContractStatus=Draft)
4. WorkroomEngagement create করে:
 - Status: `ContractPending`
 - EscrowStatus: `NotFunded`

- Milestones: Status=`FundingRequired`, EscrowStatus=`NotFunded`

5. Transaction-safe insert সব records-এর

6. Proposal update: `Status=ConvertedToProject`, `ConversionStatus=Converted`

7. Notify:

- Provider: "Workroom created" (contract confirmation-এর জন্য ready)
- Client: "Contract ready" (confirm-এর জন্য)

Frontend polling-এ: engagement resolve হলে redirect

`/dashboard/{role}/engagements/{engagementId}`।

যদি polling timeout (~30 sec): warning panel — "Your order is being set up. You can check progress in My Engagements." + button to engagements page।

Phase 5B — Manual approval path

Buyer কী দেখে:

- Result panel: "Awaiting provider approval" (Clock icon, muted tone)
- FailedConditions list দেখানো হয় verbatim (backend সরাসরি strings send করে):
 - "Instant order is disabled"
 - "Provider is unavailable or at capacity"
 - "Package capacity is full"
 - ইত্যাদি
- Button: "View my orders" → engagements page

Provider পক্ষ থেকে কী হবে:

- Provider dashboard-এ notification: "Provider approval required"
- Provider Leads page-এ proposal দেখে, review করে
- Provider `LeadsService.AcceptAsync(clientId, proposalId, r)` call করে (via approve button)
- Backend proposal update: `Status=Accepted`, background job enqueue

তারপর same as Phase 5A — buyer এসে check করলে "Pending your action" section-এ proposal দেখবে (M2b feature), "Complete your order" click করবে, দুই checkbox confirm, polling → engagement redirect।

Phase 6 — Workroom entry (contract sign)

URL: `/dashboard/{role}/engagements/{engagementId}` (client-side buyer view)

Buyer কী দেখে:

- Header: back link + service title (h1) + status chip (ContractPending)
- Left column:

- Contract panel: contract details (terms, price, delivery, revisions), "Sign contract" button
- Milestones panel: milestone cards (Status=FundingRequired, Fund button visible কিন্তু disabled)
- Files & deliverables panel
- Review panel (Completed engagement-এই সক্রিয়, এখন empty)
- Right sticky summary: Contract value, Escrow (NotFunded), Started date (null এখন), Expected end, Quick actions (Pause/Complete — Complete disabled কারণ contract sign হয়নি)

Buyer কী করে: "Sign contract" click → confirmation dialog → approve।

Backend (`SignContractAsync`), `WorkroomService.cs:195`):

- Contract status update (ClientSigned/ProviderSigned/Signed)
- যদি দুই party sign করে থাকে, Contract status = `Signed`
- Engagement status: still `ContractPending` until এই moment, তারপর `EscrowPending`-এ যায় (কোন milestone fund হয়নি এখনো)

Provider পক্ষ থেকে সমান্তরাল কাজ: SP Workroom-এ same contract sign করে (SP UI different, provider perspective)।

Both signed হলে:

- Contract Status = Signed
- Engagement Status = EscrowPending
- Fund button এখন enabled buyer-এর side-এ

Phase 7 — Fund first milestone (money enters escrow)

Buyer কী দেখে:

- Milestone card-এ "Fund escrow" button now enabled
- Amount: €X, Escrow: NotFunded
- Button click করলে confirmation dialog

Buyer কী করে: "Fund escrow" → confirm dialog → submit।

Backend (`FundMilestoneAsync`), `WorkroomService.cs:216`):

1. Precondition check করে:
 - Contract Status = Signed
 - Milestone Status = FundingRequired
2. `PaymentOperation` record বানায় (idempotency key: `escrow:{milestoneId}`)
3. **Gateway call:** `_gateway.AuthorizeEscrowAsync(key, amount, currency)` — এই মুহূর্তে `StubPaymentGateway`; ভবিষ্যতে Stripe

4. Gateway success হলে stored reference persist করে
(`PaymentOperation.GatewayReference`)
5. Transaction-safe update:
 - Milestone: (`EscrowStatus=Funded`), (`Status=Funded`)
 - Engagement: (`EscrowStatus=Funded`), (`Status=ReadyToStart`)
 - **FinancialTransaction record** create: (`Type=EscrowFunded`), (`Status=Completed`),
gross=milestone amount
6. Notify provider: "Milestone funded"

Buyer পক্ষে UI update হয় polling (30 sec) বা refresh-এ:

- Milestone status chip: "Funded"
- Engagement status: "Ready to Start"
- Fund button gone, waiting for provider to activate

Phase 8 — Provider execution (buyer waits, monitors)

এই phase-এ buyer শুধু status change দেখে; direct action করার কিছু নেই। কিন্তু বোঝা দরকার backend-এ কী হচ্ছে।

8.1 — Provider activates milestone

SP action (`ActivateMilestoneAsync`, `WorkroomService.cs:256`):

- Contract=Signed + Escrow=Funded verify
- Previous milestone Paid/Approved check (parallel না হলে)
- Milestone: (`Status=Active`), `StartDate=now`, `DueDate` computed
- Engagement: (`Status=Active`), first activation হলে `ActiveOrderCount +1` provider-এর জন্য

Buyer এই মুহূর্তে দেখে:

- Milestone chip: "In progress" (Active)
- Started date populate হয়
- Expected end date visible

8.2 — Provider submits deliverable

SP action (`SubmitDeliverableAsync`, `WorkroomService.cs:281`):

- Files scanned + Ready, ৪টা confirmation checkbox required
- (`Deliverable`) record create with version number
- Previous deliverable (revision case-এ) → Superseded
- Milestone: (`Status=ClientReviewing`) (first) বা (`Resubmitted`) (revision cycle)
- (`SubmittedAt`), (`ReviewWindowEndsAt`), (`AutoReleaseAt`) set (এই মুহূর্তে 7-day auto-release clock শুরু)
- Engagement: (`Status=MilestoneReview`)

Buyer এই মুহূর্তে দেখে (30s polling বা refresh):

- Milestone chip: "Client reviewing"
 - Files & deliverables panel-এ new delivery visible
 - Milestone card-এ ৩টা action button:
 - "Approve & release payment" (primary blue)
 - "Request revision" (outline)
 - "Open dispute" (destructive outline)
-

Phase 9 — Buyer approval decision (3 branches)

এই সবচেয়ে important decision point। Buyer ৩টা path বেছে নিতে পারে।

9.1 — Approve & release payment (happy path)

Buyer click: "Approve & release payment" → confirmation dialog → submit।

Backend (`ReleaseMilestoneAsync`), `WorkroomService.cs:703`):

1. Precondition:
 - Milestone Status = ClientReviewing OR Resubmitted
 - `DisputeOutcome != Open` (BUG-1 fix)
 - CanTransition to Paid allowed
2. **Stored** `PaymentOperation.GatewayReference` **retrieve** (BUG-3 fix; আগে fabricated string ছিল)
3. `PaymentOperation` new for release (key: `release:{milestoneId}`)
4. **Gateway call:** `_gateway.ReleaseEscrowAsync(key, storedReference, amount, currency)`
5. **Commission calculation:**
 - `commission = amount × 12%` (rounded up)
 - `net = amount - commission`
6. Transaction-safe update:
 - Milestone: `Status=Paid`, `EscrowStatus=Released`, `ApprovedAt=now`
 - Engagement: `CompletionPercentage` recomputed, `EscrowStatus=Released`
 - **FinancialTransaction** create: `Type=PaymentReleased`, `GrossAmount`, `CommissionAmount`, `NetAmount`
 - **Invoice** create: `MDL-YYYYMMDD-XXXXXX` format, tax snapshot captured
7. Trust score refresh for provider
8. Notify provider: "Payment released — €{net} available after 12% commission"

Buyer এই মুহূর্তে দেখে:

- Milestone chip: "Paid" (success tone)
- Amount shown as normal (strikethrough না, সেটা Refunded-এর জন্য)

- Approve/Revise/Dispute buttons gone
- এটা happy path — money released to provider

যদি এটাই ছিল একমাত্র milestone: engagement completion-এ যাও (Phase 11)।

যদি আরো milestones আছে: repeat from Phase 7 (Fund next milestone)।

9.2 — Request revision (correction cycle)

Buyer click: "Request revision" → revision request form (list of requested changes, "consolidated feedback" checkbox required) → submit।

Backend (`RequestRevisionAsync`, `WorkroomService.cs:330`):

- `RevisionRequest` record create
- Milestone Status → `RevisionRequested`
- Revision count check করে (`IncludedRevisionCount`) exceed করলে extra revision-এর জন্য charge — future feature)

Buyer এই মুহূর্তে দেখে:

- Milestone chip: "Revision requested"
- Waiting for provider

Provider পরে:

- `StartRevisionAsync` → Milestone Status = `RevisionInProgress`
- কাজ শেষে `SubmitDeliverableAsync` (2nd version) → Status = `Resubmitted`

Buyer আবার ৩টা option পায় (approve/revise/dispute) — same Phase 9 decision cycle।

9.3 — Open dispute (conflict path)

Buyer click: "Open dispute" → reason textarea → submit।

Backend (`OpenDisputeAsync`, `WorkroomService.cs:365`):

- Milestone Status = `Disputed`
- `DisputeOpenedAt = now`, `DisputeOutcome = Open`, `DisputeReviewEndsAt = now + review window`
- Engagement Status = `Disputed`
- Escrow frozen (auto-release sweeper skip করবে)

Buyer এই মুহূর্তে দেখে (BLOCK-2 fix-এর পরে):

- Dispute open banner: "Dispute in review"
- Milestone chip: "Disputed"
- All milestone action buttons hidden
- Copy: "Mondial support-কে review-এর জন্য পাঠানো হয়েছে"

Provider পক্ষে:

- Red error banner: "payment release blocked"
- Same milestone action-locked

Admin resolves (via `/dashboard/admin/disputes`), BLOCK-1 UI):

- **Provider-favored** → milestone `(Status=ClientReviewing)` (back to buyer approve decision), `(DisputeOutcome=ProviderFavored)`, escrow unfrozen, banner turns "info" tone (BLOCK-4)
 - **Client-favored (BUG-2 fix)** → milestone `(Status=Paid)` + `(RefundedAt=now)`, refund `FinancialTransaction`, engagement stays Active for completion
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Phase 10 — Auto-release (buyer inaction fallback)

Scenario: Provider deliverable submit করে, buyer কিছু করে না।

Hangfire background job: `(SweepTimedRulesAsync)` (`WorkroomService.cs:140`), periodic।

- সব milestones scan করে where `(AutoReleaseAt < now)` AND `(DisputeOutcome != Open)`
- Each matched milestone-এ `(ReleaseMilestoneAsync(systemId, id, autoRelease=true))` call করে
- Same commission/net calculation, same Invoice creation
- Audit event: "Milestone.AutoReleased" (System actor, not Client)
- Provider notify: "Milestone auto-released after seven days"

Buyer এই মুহূর্তে দেখে (next refresh):

- Milestone Paid (approve না করেই)
 - এটাই canon §10.7-এর 7-day auto-release invariant
-

Phase 11 — Engagement completion (all milestones done)

যখন সব milestones Paid (বা Refunded — BUG-2 semantic):

Buyer কী দেখে:

- Engagement summary card-এ "Complete engagement" button enabled
- Progress: 100%

Buyer click: "Complete engagement" → confirmation → submit।

Backend (`(CompleteEngagementAsync)`, `WorkroomService.cs:460`):

- Precondition: all milestones Paid, no open disputes
- Engagement Status = `(Completed)`, `ActualEndDate = now`
- Provider `ActiveOrderCount` -1
- Trust score refresh
- Repeat coupon check (if eligible for future services)

Buyer এই মুহূর্তে দেখে:

- Engagement status chip: "Completed"
 - Review panel এখন enabled
 - All action buttons gone
-

Phase 12 — Review + response

12.1 — Buyer submits review

Buyer কী দেখে: Review panel:

- 6 category star rating (Overall, Quality, Communication, ইত্যাদি)
- Written review textarea
- Submit button

Buyer click: submit।

Backend (`SubmitReviewAsync`), `WorkroomService.cs:481`):

- Engagement must be Completed OR Archived
- `Review` record create with ratings + text
- Trust score refresh for provider

12.2 — Provider responds

Provider পক্ষে later:

- Review notification পায়
 - Response textarea via UI
 - `RespondToReviewAsync` — এই action **engagement termination-এর পরেও চলে** (এটাই `DeliveriesPanel readOnly` bug-এর discovery ছিল)
-

Phase 13 — Provider payout (separate flow, later)

Buyer flow এখানেই শেষ। কিন্তু provider পক্ষে payment cycle সম্পূর্ণ হতে আরো একটা step:

- Net amount (gross - 12% commission) provider account-এ available balance-এ যোগ হয়
- Provider `PayoutRequest` create করে UI থেকে (Earnings page)
- Backend gateway call → `StubPayoutGateway` (future: Stripe Connect)
- Success হলে `FinancialTransaction` (`Type=PayoutCompleted`)
- Provider bank/card-এ money reach করে (real gateway-এ)

Buyer এই phase-এ কোনো interaction নেই।

Money flow summary (100 EUR order example)

```
graph TD
    A[Buyer pays 100 EUR] --> B[Phase 7: FundMilestone]
    B --> C[Escrow authorized: 100 EUR]
    C --> D[FinancialTransaction: EscrowFunded, Gross=100]
    D --> E["(work happens, provider submits, buyer approves)"]
    E --> F[Phase 9.1: ReleaseMilestone]
    F --> G[Escrow released: 100 EUR]
    G --> H["Commission (12%): 12 EUR → Mondial"]
    H --> I["Net: 88 EUR → provider available balance"]
    I --> J["FinancialTransaction: PaymentReleased, Gross=100, Commission=12, Net=88"]
    J --> K["Invoice: MDL-20260802-XXXXXX"]
    K --> L["(provider requests payout, later)"]
    L --> M[Phase 13: Payout]
    M --> N[Provider account: +88 EUR]
    N --> O[FinancialTransaction: PayoutCompleted, Gross=88]
```

States quick reference

EngagementStatus lifecycle (happy path):

```
ContractPending → EscrowPending → ReadyToStart → Active → MilestoneReview →
ReadyToStart (next milestone) → ... → FinalDelivery → Completed
```

MilestoneStatus lifecycle (happy path):

```
FundingRequired → Funded → Active → ClientReviewing → Paid
                                     |
                                     └─ (revision) → RevisionRequested →
RevisionInProgress → Resubmitted → Paid
                                     |
                                     └─ (dispute) → Disputed → [admin
resolves] → ClientReviewing (prov-fav) OR Paid+RefundedAt (client-fav)
```

EscrowStatus lifecycle:

NotFunded → Funded → Released (via approve/auto-release) OR Refunded (via client-fav dispute)

Real-world timing expectations

- **Phase 3 (order wizard):** buyer time-dependent, typically 2-5 min
- **Phase 4 (backend decision):** instant, <500ms
- **Phase 5A (auto-accept):** proposal conversion ~1-5 sec background, frontend polling picks up in 15-30 sec
- **Phase 5B (manual approval):** provider time-dependent, minutes to hours
- **Phase 6 (contract sign):** both parties needed, minutes to hours
- **Phase 7 (fund):** instant once buyer decides, gateway ~1-2 sec
- **Phase 8 (work execution):** days to weeks depending on scope
- **Phase 9 (review):** buyer time-dependent, or 7-day auto-release
- **Phase 10 (auto-release):** exactly 7 days after `AutoReleaseAt`
- **Phase 11 (completion):** instant on button click
- **Phase 13 (payout):** provider time-dependent, gateway hours to days in real gateway

Currently unsupported / limitations (documented gaps)

- **File download absent** — buyer delivery listed দেখে, open করতে পারে না (highest user-value gap, backend endpoint missing)
- `completionCriteria` + `autoReleaseAt` **buyer visibility** — approve করে unseen criteria-এর against, unknown 7-day clock
- **Real payment gateway** — এখন StubPaymentGateway, M8-এ Stripe swap হবে (BUG-3 fix সেই swap-এর prerequisite)
- **Chat/Message provider** — button আছে UI-এ but stub, actual chat backend built না
- **Provider display name** — engagement list card + detail header-এ absent (backend DTO gap)